RELIABILITY OF DELAYED BROWSER PERFORMANCE DIAGNOSIS

Final Report

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N. W. L. U. R. D. Nanayakkara

EEY6189 Research Methodology and Project Identification
(Software)

Faculty of Engineering Technology
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ABSTRACT

A detailed browser profile collected after a performance alert describes a new execution. It may therefore explain the follow-up visit without identifying the cause of the original regression. This report proposes a controlled longitudinal study of that diagnostic gap. Scheduled synthetic visits to controlled website fixtures will be used to compare lightweight original-visit attribution, detailed follow-up profiling, bounded original-event capture and periodic detailed capture under common resource allowances. Always-rich capture will be assessed under a separate reference allowance. Fault activity will be checked independently on both original and follow-up visits, with persistent, transient and replaced causes distinguished. The primary comparison will be the difference in correct original-cause attribution between retained evidence and follow-up profiling for intermittent faults. All verified eligible original episodes will be included, including missed alerts, failed captures, late diagnoses, abstentions and exhausted policies. Resource use, diagnosis delay and misleading attribution will also be measured. A finite feasibility pilot with two fixtures is specified before a separate main-study protocol and analysis freeze. Neither the pilot nor the main study has been run; no experimental findings or hypothesis results are reported. The intended contribution is a bounded account of when delayed profiling can explain an earlier episode and when original-visit evidence is needed, subject to verified capture feasibility, complete cost accounting and the limits of the tested conditions.

Keywords: website performance; browser profiling; delayed diagnosis; original-visit evidence; longitudinal measurement; fault attribution

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LIST OF SYMBOLS AND ABBREVIATIONS

Abbreviation or symbol	Meaning
API	Application programming interface
CDP	Chrome DevTools Protocol
CLS	Cumulative Layout Shift
CPU	Central processing unit
DOM	Document Object Model
FCP	First Contentful Paint
FP-0.1	Finite pilot protocol, version 0.1
GiB	Gibibyte (binary storage unit)
H_0	Null hypothesis
H_1	Research hypothesis
INP	Interaction to Next Paint
LCP	Largest Contentful Paint
LoAF	Long Animation Frames
MAD	Median absolute deviation
MiB	Mebibyte (binary storage unit)
OS	Operating system
RSS	Resident set size
RUM	Real-user monitoring
SI	Speed Index
TBT	Total Blocking Time
TTFB	Time to First Byte
TTI	Time to Interactive

01. INTRODUCTION

1.1 Background and Motivation

Website performance is commonly examined through repeated browser visits. A lightweight measurement can identify that a visit was slow or that a metric changed, while detailed profiling can expose resource timing, script activity, or a page's dependency path. These observations answer different questions. A metric describes an outcome under defined measurement conditions. Attribution fields offer clues about that visit, while detailed instrumentation can expose additional execution activity. The information and cost depend on the capture configuration (GoogleChrome/web-vitals contributors n.d.; Netravali et al. 2018).

The difficulty is that a follow-up visit is a new execution. Suppose an original visit is slow because an image response is delayed, but the delay disappears before the analyst opens a rich profile. The follow-up profile may be internally accurate and still fail to explain the earlier visit. Conversely, if the fault remains active, a later profile may reproduce useful evidence. This contrast motivates a test of evidence transfer across visits; it does not establish that reruns generally fail. Vesper describes a two-pass browser measurement design and discusses cross-load variation, while its tested corpus did not show different functional state under its own setup (Netravali et al. 2018). The proposed study will preserve visit identity so that the original and follow-up observations cannot be mistaken for one another.

The study is planned as repeated scheduled synthetic measurement on controlled, owned or authorized fixtures. "Longitudinal" here means a sequence of planned visits under recorded conditions; it does not mean continuous collection from every real user. A site-specific history can help describe its usual variation, but a historical baseline alone does not prove the cause of a particular anomalous visit. Detecting an unusual episode and attributing its cause are therefore separate tasks.

1.2 Problem Statement

The problem is whether a detailed profile collected after an alert can reliably identify the independently verified cause of the original website performance regression. A policy that waits for a later visit may spend less on every visit, but it depends on relevant evidence still being observable. A policy that retains original-visit evidence may preserve event-specific clues, but its feasible event coverage, lifecycle, data loss, and collection cost must be established. The principal strategies will therefore be compared under common resource allowances and a diagnosis deadline. Original collection and failed or uninformative attempts will be charged. Always-rich capture will have a separate allowance and will provide a coverage and cost reference.

The empirical question is deliberately narrower than a claim that one profiler is universally superior. It asks whether later evidence identifies the earlier cause for a stated set of fixtures, fault mechanisms, browser states, and delays. An independent fault record is required because a profiler's trace is evidence produced by the measurement system, not ground truth by itself. The unresolved research issue is the reliability boundary across persistent and intermittent faults, not the invention of adaptive profiling, website histories, browser tracing, or original-event retention (Miller et al. 1995; Grassi et al. 2021; Hantke et al. 2025; Zhang et al. 2023).

1.3 Research Question and Aim

Research question. Within a fixed monitoring budget, how reliably can detailed profiling during a follow-up browser visit identify the cause of an earlier website performance regression, compared with diagnostic evidence saved during the original visit?

Aim. Assess the reliability, diagnostic coverage and total monitoring cost of delayed browser profiling compared with retaining diagnostic evidence from the original visit.

Reliability will be assessed as correct attribution of the original fault class within a shared diagnosis deadline, with every independently verified eligible episode accounted for. "Fixed budget" will refer to a published resource vector and allocation rule, not to an unexplained single score. The resource dimensions will be reported separately.

1.4 Hypotheses

Research hypothesis (H_1). For intermittent website performance regressions, retaining diagnostic evidence from the original browser visit leads to a higher proportion of correctly attributed regression episodes than detailed profiling during a follow-up visit, under matched monitoring budgets and the same diagnosis deadline.

Null hypothesis (H_0). Under the same conditions, retaining diagnostic evidence from the original browser visit does not increase the proportion of correctly attributed intermittent regression episodes.

Correct attribution means identifying the independently verified fault class responsible for the *original* episode. Missed alerts, failed follow-up visits, unavailable evidence, and abstentions remain in the denominator. Total monitoring cost includes original collection, retained evidence, confirmations, and follow-up visits. These are hypotheses to test, not expected or observed results. Fault persistence is the principal factor; follow-up delay and browser-state changes are secondary factors within the study's resource limits.

1.5 Objectives

The study has five objectives:

1. **Define a repeatable measurement protocol** that records browser conditions, metric definitions and evidence provenance, and clearly separates original visits from diagnostic follow-up visits.
2. **Build a longitudinal, website-specific evaluation dataset** with stable periods, persistent and intermittent regressions, and different causes that produce similar symptoms. Use independently verified fault labels.
3. **Implement and compare feasible evidence-collection strategies**, including lightweight original-visit attribution, retained original-event evidence, detailed follow-up profiling and periodic detailed capture.
4. **Evaluate diagnostic effectiveness and total monitoring cost** by measuring correct original-cause attribution, missed episodes, misleading diagnoses, abstentions, diagnosis delay, instrumentation effects and capture failures.
5. **Identify when delayed diagnosis is reliable**, focusing on fault persistence and assessing the effects of follow-up delay and browser-state changes.

Objective 3 depends on establishing that bounded original-event capture can retain useful diagnostic information within its limits. If this cannot be achieved, the comparison will be narrowed or revised before the main-study protocol is frozen.

1.6 Scope and Contribution Boundaries

The initial scope is a controlled experiment on a small set of owned or authorized website fixtures, with scheduled synthetic Chromium visits and known injected or configured faults. The proposed feasibility version FP-0.1 includes two fixture patterns and a bounded set of persistent, transient, replacement, stable, and nuisance conditions; its caps and progression rules are proposed engineering choices for feasibility assessment. A later live-site sample may test operational robustness only if explicitly authorized. Unexplained slowdowns on live sites cannot supply verified cause labels.

The intended contribution is an empirical comparison of original-visit and follow-up evidence for attribution of the original cause. It includes diagnostic coverage, failures, total cost and the conditions under which a later diagnosis remains valid. Adaptive instrumentation, browser profiling, recorded-execution analysis and bounded event retention have already been studied (Miller et al. 1995; Netravali et al. 2018; Hantke et al. 2025; Zhang et al. 2023). The proposed contribution is therefore limited to the specific comparison and evidence described here. Its novelty remains provisional because the available literature review does not establish the absence of an equivalent study.

The research also supports a broader browser-profiling engineering responsibility. The planned software deliverables include a profiler service and application programming interface (API), together with seven performance views: Live Performance Test, Performance Overview, Interactive Performance Timeline, Network Requests, CPU and Main Thread Analysis, Memory and DOM Analysis, and Performance Issue Details. Metric, network, main-thread, memory and timeline outputs will be accompanied by evaluation and unit tests for URL validation, timing calculations and score generation. The pilot uses a smaller endpoint and fault set to examine diagnostic reliability; it does not remove these engineering duties. Comparisons with Chrome DevTools or Lighthouse will assess measurement consistency where definitions and conditions match. Since these tools share browser evidence, agreement will not establish independent fault truth.

1.7 Report Organization

Chapter 2 reviews metric definitions, browser capture, repeated measurement and related diagnosis systems. Chapter 3 presents the proposed experiment, independent fault checks, strategies, costs and analysis. Chapter 4 describes the research sequence and its dependencies. Chapter 5 brings together the intended contribution and the limits on the conclusions that can be drawn at this stage.

02. LITERATURE REVIEW

Browser performance diagnosis draws on metric definitions, instrumentation, repeated measurement and methods for retaining or selecting execution evidence. Existing systems establish several combinations of these mechanisms. The remaining question is whether evidence from a later browser visit can correctly identify an earlier cause under the proposed budget, deadline and independent labelling rules.

2.1 Browser Performance Metrics and Semantics

A performance endpoint and a diagnostic signal serve different purposes. Largest Contentful Paint (LCP) records the render timing of an eligible largest content element; it does not mean that a page is complete or that a particular resource caused a delay. Interaction to Next Paint (INP) summarizes responsiveness across qualifying interactions and depends on an input actually occurring; it cannot be inferred from a load-only visit. Cumulative Layout Shift (CLS) aggregates eligible layout instability according to session-window rules, rather than summing every shift over a page's entire lifetime. These definitions are provided by the metric specifications, guidance and implementation documentation (World Wide Web Consortium 2026a; *Interaction to Next Paint (INP) | web.dev* n.d.; *Layout Instability API* n.d.; *Cumulative Layout Shift (CLS) | Articles* n.d.; GoogleChrome/web-vitals contributors n.d.).

Other metrics describe different aspects of performance. First Contentful Paint (FCP) describes an earlier rendering milestone, Total Blocking Time (TBT) is a lab diagnostic over a specified interval, Speed Index (SI) is based on visual progression, and client-observed Time to First Byte (TTFB) describes time to the first response byte (World Wide Web Consortium 2026c; *Total Blocking Time | Lighthouse* n.d.; *Speed Index | Lighthouse* n.d.; World Wide Web Consortium 2026b). For a navigation, the start point matters: an interval from navigation start to `responseStart` includes work before the request, while an interval from `requestStart` has a narrower scope. The exact navigation formula, redirect handling and cache context will be fixed in the measurement contract. TTFB is not an isolated measure of server CPU time (World Wide Web Consortium 2026b; *HTTP* n.d.). Subresource request phases will be recorded separately using Resource Timing, whose fields and access restrictions have their own definitions (*Resource Timing* n.d.). Event Timing and the Layout Instability API describe underlying browser observations; they are not interchangeable with the final aggregate reported to users (*Event Timing API* n.d.; *Layout Instability API* n.d.). The cited specification editions include the LCP Working Draft of 26 August 2026, Paint Timing Working Draft of 2 October 2026 and Navigation Timing Level 2 Working Draft of 1 September 2026. These definitions still require a fixed implementation and measurement contract. The proposed evaluation will begin with pilot LCP as its

primary endpoint; INP and CLS remain exploratory unless a qualifying journey and finalization rule are validated.

Each metric will be recorded with its definition and version, eligible navigation or interaction, observation and finalization window, browser context, export behavior, scope and missingness reason. A missing interaction or unsupported API is not a zero. Metric attribution can offer a candidate element or phase without proving the injected fault caused the result. The web-vitals v5.1.0 attribution documentation establishes a legitimate source of original-visit diagnostic fields; it is a reference release, not the selected runtime build or a cause oracle (GoogleChrome/web-vitals contributors n.d.). FP-0.1 proposes LCP as the pilot endpoint and a median plus max(100 ms, 3 times unscaled median absolute deviation) alert threshold. These are project choices to qualify and lock before comparison, not externally validated standards. Exact runtime field mapping and finalization still require conformance checks; main-study thresholds and aggregation remain to be frozen separately.

2.2 Browser Instrumentation and Measurement Effects

A summary metric retains less execution detail than many forms of trace collection, but the difference depends on the fields actually recorded. Browser application programming interfaces (APIs), including the Performance Timeline, offer typed entries and observer/buffer mechanisms; Event Timing, Long Tasks, and Long Animation Frames (LoAF) expose particular classes of activity. The Chrome DevTools Protocol (CDP) tracing domain exposes trace events subject to implementation and lifecycle details (World Wide Web Consortium 2025; *Event Timing API* n.d.; *Long Tasks API* n.d.; *Long Animation Frames API* n.d.; Chromium authors n.d.). The cited Performance Timeline edition is the Candidate Recommendation Draft of 21 May 2025. Its buffering mechanisms are finite and entry-type specific; runtime support will be checked rather than inferred from the specification. These sources justify specifying event types, timestamps, ownership, observer start time, buffer capacity, export, truncation and loss. They do not by themselves establish that the collected fields distinguish the study's fault classes or that collection has negligible cost.

In this study, "lightweight" and "detailed" will identify capture contracts. Each contract will specify the collector, entries or profiles retained, start and stop times, retention limits, export size, loss handling and resource costs. This is particularly important for a retained-original-event arm. A bounded buffer can be useful only if it starts in time and contains the needed discriminator before overflow, page termination, or export. Hindsight demonstrates finite-horizon original-event retention and retrieval for distributed systems; it motivates a feasibility test but does not prescribe a browser buffer size (Zhang et al. 2023).

Instrumentation also changes what is measured. Mytkowicz and colleagues show that apparently innocuous measurement choices can produce misleading performance data in systems experiments (Mytkowicz et al. 2009). That work motivates calibra-

tion; it does not quantify the overhead of the browser collectors proposed here. Minimally observed and profiled executions will therefore be compared in matched blocks. Changes in timing, central processing unit (CPU) use, failure and fault activity will be measured, and failed or truncated captures will be retained. A browser trace is an observation generated under instrumentation, not independent evidence that a fault was active.

2.3 Repeated Measurement and Experimental Validity

Repeated visits support longitudinal comparison only when their conditions are recorded. Cache state, browser profile, server state, network and CPU conditions, session data, and workload order can alter a visit. WProf shows the value of decomposing browser load behavior and conditioning interpretation on workload and state; its historical Chromium results do not establish a modern repeat count or a protocol for this study (Wang et al. 2013). Vesper's two-pass approach provides a browser-specific example of lighter measurement after heavier discovery. Its authors discuss possible differences between loads, but no different functional state was observed in their tested corpus. Vesper places heavy discovery before lighter measurement; the present study instead examines richer profiling after a lightweight original visit (Netravali et al. 2018).

Web Execution Bundles (WebREC) provide a particularly relevant contrast: archiving an execution can support later analysis of the recorded visit without treating a new visit as equivalent. This is prior art for preserving original-execution evidence. It does not establish that the pilot's browser event buffer is feasible, covers the chosen performance endpoints, or is cheaper than a fresh diagnostic visit (Hantke et al. 2025). The cited arXiv v4 manuscript supports this distinction; targeted passages in the final proceedings paper corroborate it without establishing complete equivalence between editions. Lab measurements also have a different target from real-user monitoring (RUM); controlled interventions can support causal testing when their execution and intended effect are independently checked under declared conditions. A configured fault flag alone is insufficient. Such a corpus does not establish production prevalence or a field percentile.

Randomized order, explicit browser and fixture resets, repeated stable histories, chronological splits and records of actual state will be used to improve comparability. A new browser context isolates browser storage, but does not reset every server, host, or shared cache condition. A site-specific baseline can characterize earlier observations; it cannot alone determine why a particular visit was anomalous. Repeated visits are not independent websites; conclusions will be limited by the tested fixtures and conditions.

2.4 Related Diagnosis and Monitoring Systems

Table 2.1 summarizes the closest mechanisms and the boundary relevant to the proposed question. The comparison distinguishes established mechanisms from the evidence still required for this browser-specific study.

Table 2.1: Closest prior systems and the bounded question left for this study.

System/source	Established contribution	Limit relevant to this study
Paradyn (Miller et al. 1995)	Hypothesis-guided dynamic instrumentation uses an explicit collection-cost threshold and discusses reruns for short programs.	The cited project manuscript is not a browser study; exact equivalence to the journal-edited text remains unresolved.
Spectroscope (Sambasivan et al. 2011)	Diagnoses performance changes by comparing healthy and problem request flows in distributed systems.	Uses observed request-flow evidence; it does not test a new browser visit as a substitute for the original episode.
Shahedi et al. (Shahedi et al. 2024)	The 2024 arXiv v1 manuscript studies selective native function tracing for performance modelling and regression detection.	Compares program versions and inputs; it does not establish recovery of an earlier transient browser cause. The final journal text has not been verified.
WProf (Wang et al. 2013)	Browser dependency analysis locates load-path bottlenecks and shows the relevance of workload and cache conditions.	Historical browser setup; it does not test whether a later profile explains an earlier intermittent fault.
Vesper (Netravali et al. 2018)	Uses a heavy discovery pass and lighter measurement pass for browser interactivity.	Cross-load differences are a motivation, not a universal failure claim; its historical readiness metric is not current INP.
WebREC (Hantke et al. 2025)	Records web executions for later analysis and supports reasoning from original-visit evidence.	Endpoint coverage and feasibility of a bounded, low-cost browser buffer remain to be tested for this study.
Hindsight (Zhang et al. 2023)	Retains bounded original-event evidence for later retrieval after rare distributed-system events.	Distributed-system design; browser event completeness, lifecycle loss and total cost are unestablished.

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System/source	Established contribution	Limit relevant to this study
VisibleV8 (Jueckstock and Kapravelos 2019)	Demonstrates in-browser JavaScript monitoring with overhead that depends on capture configuration.	Historical Chromium measurements do not compare delayed diagnosis with original-episode labels.
LMAT (Darvishi et al. 2026)	Describes an adaptive host-level tracing loop that reacts to performance deviations.	Full publisher text was unavailable; host-level evidence does not establish browser cross-visit attribution.
WMF (Grassi et al. 2021)	Studies website performance anomalies using sparse histories and controlled/synthetic cases.	Anomaly detection or reconstruction is distinct from verified attribution of the cause of a specific earlier browser visit.
Adaptive trace sampling (Mertz and Nunes 2023)	Selects execution-trace samples under resource constraints.	Representative sampling in its tested systems does not establish that a transient browser-fault discriminator survives selection.

The Paradyn comparison uses the cited project manuscript, whose exact correspondence to the edited journal article remains unverified. Shahedi et al. are cited specifically as arXiv:2411.17548v1, submitted in 2024; the later journal association does not verify the final text. For LMAT, accessible indexed passages and author material support broad host-level adaptive tracing, but conflicting descriptions of pre-trigger buffering and post-trigger recording remain unresolved. No exact controller behavior, loss rate, accuracy or overhead is adopted from that evidence. The adaptive-sampling comparison concerns Mertz and Nunes's arXiv v1 manuscript; equivalence to the journal text has not been established. The introduction and background of Argus also describe causal diagnosis of a Chrome/macOS responsiveness problem, although its evaluation and complete-cost equivalence have not been appraised here (Weng et al. 2021).

Hindsight's trace-retention lifecycle (Figure 2.1) illustrates bounded retention and later retrieval in a distributed system. Browser implementation and capture feasibility remain questions for the proposed study.

2.5 Research Gap and Study Positioning

Existing work establishes these mechanisms and several combinations of them: current browser metrics and attribution, browser instrumentation, selective or adaptive tracing, site-specific histories, recorded execution, and bounded retention. The proposed contribution is the narrower empirical comparison: whether delayed rich profil-

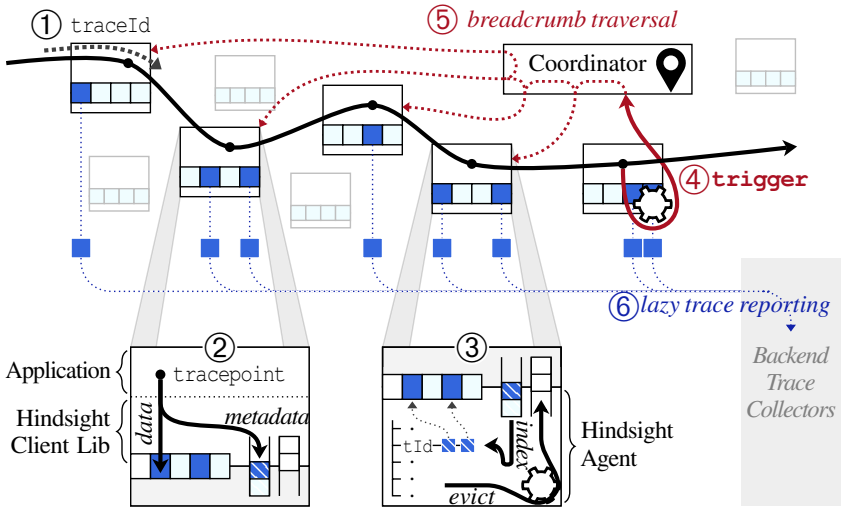


Figure 2.1: Hindsight's original-event trace-retention lifecycle, reproduced as a crop from Figure 2 (PDF p. 6; printed p. 325) to illustrate distributed-systems prior art. It is not a browser design or a result of the proposed study (Zhang et al. 2023).

ing correctly attributes an earlier browser regression episode compared with feasible evidence saved on the original visit, under common budgets and a fixed deadline, with independent verification of fault activity on each relevant visit.

The proposed gap remains a research question rather than proof that no equivalent exists. Existing lightweight attribution may already classify every selected cause, leaving little value for richer collection in this fault set. Closely related systems may also narrow the contribution after further comparison. The study will therefore assess added diagnostic value rather than assume an advantage for retained evidence. If a useful bounded original-event record cannot be collected, the comparison will be revised before the main study.

Fault persistence will be the principal condition examined, with follow-up delay and browser state treated as secondary factors. A correct explanation of the follow-up visit will be distinguished from a correct explanation of the original cause. Failures and abstentions will remain visible, and time, CPU, memory and stored bytes will be reported separately. These choices make the comparison testable while limiting conclusions to the browser, fixtures, faults and observation schedule actually evaluated.

03. RESEARCH METHODOLOGY

3.1 Research Design and Status

A controlled longitudinal experiment is proposed to compare delayed browser profiling with evidence retained during the original visit. Neither the finite feasibility pilot, FP-0.1, nor the main study has been run. The procedures and numerical settings in this chapter are proposed protocol choices. Actual host capacity, software builds, capture fields and classifier mappings remain to be verified before collection.

The main unit of evaluation will be an *original regression episode*. Each affected scheduled original visit will receive an episode identifier and will be linked to any diagnostic follow-up. Visits affected by one continuing fault occurrence will retain that shared occurrence identifier. They will not be treated as independent sites. Each strategy will receive a separate execution within a matched block; matching will align the assigned fixture, scenario and repetition, but cannot make nondeterministic executions identical (Kalibera and Jones 2013; Kalibera and Jones 2020; Jueckstock, Sarker, et al. 2021).

The primary comparison will be retained original-event evidence (C) minus detailed follow-up profiling (B) for intermittent faults under common resource allowances and a diagnosis deadline. Objectives O1–O5 will be addressed through the measurement contract, labelled dataset, strategy comparison, outcome analysis and assessment of reliability conditions. The pilot will assess feasibility. The main-study sample, target population, weighting, practical effect criterion, uncertainty method and stopping rule will be fixed separately after the pilot.

3.2 O1: Measurement and Visit Contract

3.2.1 Environment and Visit Boundaries

Chromium, automation, profiler and metric-library versions will be fixed and recorded with checksums. The host, operating system (OS), background load, viewport, CPU and network settings, browser process and profile, cache and session state, and service-worker policy will be recorded. Intended and observed conditions will be stored separately. A fresh browser context isolates browser storage, but does not establish that server, host, Domain Name System or all cache state has been reset (Microsoft / Playwright contributors n.d.; *Emulation - DevTools Protocol* n.d.; *Chrome DevTools Protocol* n.d.).

FP-0.1 is limited to two owned localhost fixtures, one headless Chromium build, one automation stack and one host. A desktop viewport of 1365 by 768 pixels, device

scale factor 1, no network throttling and CPU rate 1 are specified, except for the nuisance condition. Service workers, authentication, human participants, mobile devices, live websites and multiple browsers are outside the pilot. These limits apply to feasibility work, not to the complete engineering deliverables.

The visit clock will start at attempted browser-process launch. Collection will start before fixture code executes. Each visit will be observed for 15 seconds from navigation start, with a safe button input attempted at 5 seconds. An unavailable input will be recorded as a failure. Collectors will be stopped at 15 seconds and flushed while the page remains available. Startup, observation, export and cleanup must fit within the 30-second visit limit. Original and follow-up visits will have separate identifiers, local time origins and explicitly aligned clocks.

3.2.2 Metrics and Capture Definitions

Largest Contentful Paint (LCP), measured in milliseconds, will be the primary pilot symptom. Its candidate, input cutoff and finalization rule will be fixed to the selected implementation. Client-observed navigation and resource timings and supported attribution will also be recorded. The navigation TTFB start point, timestamp formula, redirects and cache scope will be specified before collection; subresource intervals will remain separate. Interaction to Next Paint (INP) and Cumulative Layout Shift (CLS) will be exploratory checks of availability and lifecycle handling. Missing or unsupported values will receive reason codes, not zeros. Total Blocking Time (TBT), composite scores, memory-growth diagnosis and full metric validation are outside the pilot endpoint. TBT will not be treated as a substitute for INP, and historical Time to Interactive (TTI) will not be restored as an endpoint (*Largest Contentful Paint | Lighthouse* n.d.; *Total Blocking Time | Lighthouse* n.d.; *Time to Interactive | Lighthouse* n.d.; GoogleChrome/web-vitals contributors n.d.; World Wide Web Consortium 2026b; *Resource Timing* n.d.).

Each capture contract will identify the collector, event types and fields, frame and process scope, start and stop times, sampling or thresholds, buffer limits, loss handling, serialization and costs. The candidate rich contract will combine Chrome DevTools Protocol (CDP) Network metadata with main-thread, paint, layout and script trace events. A sampled CPU profile will be included only if the selected build supports the chosen configuration. Response bodies, video and heap snapshots will be excluded. The exact categories and fields will be qualified before the contract is treated as implemented. A warning-free trace will not be assumed complete (World Wide Web Consortium 2025; Chromium authors n.d.; Web Platform Incubator Community Group n.d.; *Performance panel* n.d.).

3.2.3 Conformance and Instrumentation Effects

Conformance checks will link known events to the expected visit, timestamp, resource or script. Metric definitions, callback finalization, export and loss signals will

Table 3.1: Minimum measurement-contract fields. Interface documentation identifies candidate evidence; runtime qualification will establish the implemented contract (World Wide Web Consortium 2025; GoogleChrome/web-vitals contributors n.d.; Chromium authors n.d.).

Record	Required fields
Environment and journey	Builds and hashes; host, viewport and profile; intended and actual state; fixture and navigation; input, observation and finalization times.
Evidence and missingness	Visit identifier and role; collector, categories and scope; drop and eviction intervals; export and hash; unsupported, absent, failed, timed-out or lost reason codes.
Resource cost	Elapsed time, CPU time, resident memory and bytes by stage; attempts, follow-ups, failures and remaining allowance.

be checked on simple fixtures. Comparisons with Chrome DevTools or Lighthouse will assess consistency where their definitions and conditions match. Since these tools share Chromium evidence, agreement will not supply an independent fault oracle (*[VIM3] 2.13 measurement accuracy* n.d.; *[VIM3] 2.18 measurement bias* n.d.; *Performance panel* n.d.; *What's new in DevTools, Chrome 143 | Blog* n.d.).

A minimally observed contract, M, will contain a minimal LCP observer, harness timing and the common hidden verification probe. Its measurement cost is nonzero. Separate M, A, C and rich executions will be compared in randomized calibration blocks to assess timing changes, failures and altered fault activity. The proposed calibration has six blocks per fixture: four stable, one R and one J. The stable blocks will provide a small-sample LCP perturbation summary; the fault blocks will check activity under every contract. The resulting measurement contract, conformance matrix and state-reset procedure will be versioned. Unsupported endpoints will be excluded from the main contract unless their failures are resolved.

3.3 O2: Longitudinal Dataset and Fault Oracle

3.3.1 Fixtures and Fault Scenarios

Fixture F1 will be an image-led page with a designated LCP candidate. F2 will use a small script to construct its designated LCP content and will have a different Document Object Model (DOM) and loading structure. Both will provide a safe button input and support both fault classes. External dependencies, cross-origin frames and background polling will be excluded. URLs, element identifiers and content structure will remain unchanged across fault conditions. Policy-visible visit identifiers will be opaque and will not encode fault names or schedules.

Fault R will delay a designated critical response by an additional 500 ms. Its actual execution will be checked using evaluator-only request identifiers and start/end logs. Fault J will introduce approximately 500 ms of main-thread work before the designated LCP content can paint. A common evaluator probe will record its entry, exit and elapsed work; the probe's cost will be measured. Probe records will be sent through a separate evaluator channel that policy processes cannot read. Policy inputs and saved evidence will be audited for accidental fault-marker leakage.

Both faults are intended to produce similar loading symptoms through different mechanisms. Qualification will establish whether each fault executes and produces the intended effect under minimally observed conditions. Performance mutation testing provides a precedent for controlled performance changes, but its native-program experiments do not validate these browser faults or supply transferable repetition counts (Delgado-Pérez et al. 2021). A configuration flag or a rich trace alone will not be accepted as proof of fault activity.

Table 3.2: Proposed FP-0.1 scenarios. Follow-up activity will be checked independently on each visit.

Scenario	Original visit	B follow-up, if alerted
Stable	No fault; CPU rate 1	No fault; CPU rate 1
Nuisance	No software fault; CPU rate 2	Same CPU setting; no fault
Persistent R / J	R / J	Same assigned class
Transient R / J	R / J	No fault
Replaced R / J	R / J	J / R respectively

The paired rows in Table 3.2 represent eight scenarios in total. Persistence will be controlled by visit role, with a separately reset occurrence for each arm. This tests known conditions; it does not estimate natural fault lifetimes. Repeated slowness on a follow-up will not establish that the original cause recurred.

3.3.2 Stable History, Alert and Diagnosis Rules

For each fixture, the first 20 stable A visits will train the baseline and the next 20 will be used for validation. The proposed alert threshold is the median stable LCP plus the greater of 100 ms and three times the unscaled median absolute deviation (MAD). This operational threshold is neither a significance test nor a field performance threshold. It will be locked after qualification and will not be updated during the comparison.

A small rule-based classifier will be fixed after qualification. Its candidate discriminators are a critical-resource phase at least 200 ms above its stable median and at least 200 ms of attributable pre-paint main-thread work. Availability, overlap and ambiguity will be checked, and the final field-to-class mapping will be saved before com-

parison. Supported attribution available to A will be retained even when it already distinguishes the faults. If both explanations are supported, neither is supported or the evidence is incomplete, the classifier will abstain.

All arms will issue cause claims only after the common alert. The vocabulary will be R, J or abstain; controls will also allow "no supported target fault." For B, supported original evidence will take priority. A class supported only by the follow-up will be marked as an inference about the original cause from a later visit. Conflicting original and follow-up evidence will be flagged rather than overwritten.

3.3.3 Independent Labels, Eligibility and Provenance

An evaluator-only intervention manifest and per-visit activity and applicability checks will define the labels. An eligible original opportunity will be a scheduled visit in a prequalified fault case whose fault execution and declared journey/endpoint applicability are independently confirmed. Eligibility will not depend on an alert, a slow observed value, metric availability or successful capture. A fault-induced timeout, missing policy metric or missed alert will not remove an otherwise eligible opportunity.

Cases with unresolved activity or cause will remain in the assignment ledger with an unresolved-label status, outside known-cause scoring. The same eligibility rule will be applied to all arms. Predictions will be frozen before hidden labels are joined for evaluation. Intended interventions, verified execution and policy diagnoses will remain separate records. This preserves the distinction between what was scheduled, what occurred and what the strategy inferred.

Table 3.3: Linked dataset records. Explicit units, state and missingness support reproducibility; the fault oracle is a study-specific design (Demir et al. 2022; Jueckstock, Sarker, et al. 2021; Delgado-Pérez et al. 2021).

Record	Minimum contents
Site and configuration	Template, deployment, journey, versions, host and profile; intended and observed state.
Visit and episode	Visit, parent episode, block, occurrence and session identifiers; role and timings; class, onset and duration; activity, persistence, applicability and label status.
Evidence and diagnosis	Raw file references and hashes; capture contract, window and loss; prediction or abstention, diagnosis time and evidence available to the policy.
Cost and attempt	Resource dimensions, stage, outcome and failure codes, remaining allowance, follow-ups and evaluator-only cost.

An all-attempt record will separate assigned, attempted, verified eligible, known ineligible, unresolved-label and globally unattempted opportunities. Raw evidence,

derived metrics and labels will be stored separately. For the main study, historical training, tuning and later test periods will be separated, and distinct fixture templates or site instances will be reserved before tuning. Pilot and repair observations will be excluded from the confirmatory dataset.

3.4 O3: Capture Strategies and Allocation

3.4.1 Evidence Available to Each Strategy

All strategies will share the lightweight measurements, baseline, alert rule, fault vocabulary, scheduled original opportunities and diagnosis deadline. Each will receive its own execution. Arm order will be randomized within scenario blocks, the specified state will be reset and only one browser workload will run at a time. Blocking controls assigned factors but does not remove every source of carryover (*5.3.3.2. Randomized block designs* n.d.; Kalibera and Jones 2013).

Figure 3.1 shows how evidence and labels will be separated. A locked diagnosis will be joined to the verified original label only at evaluation.

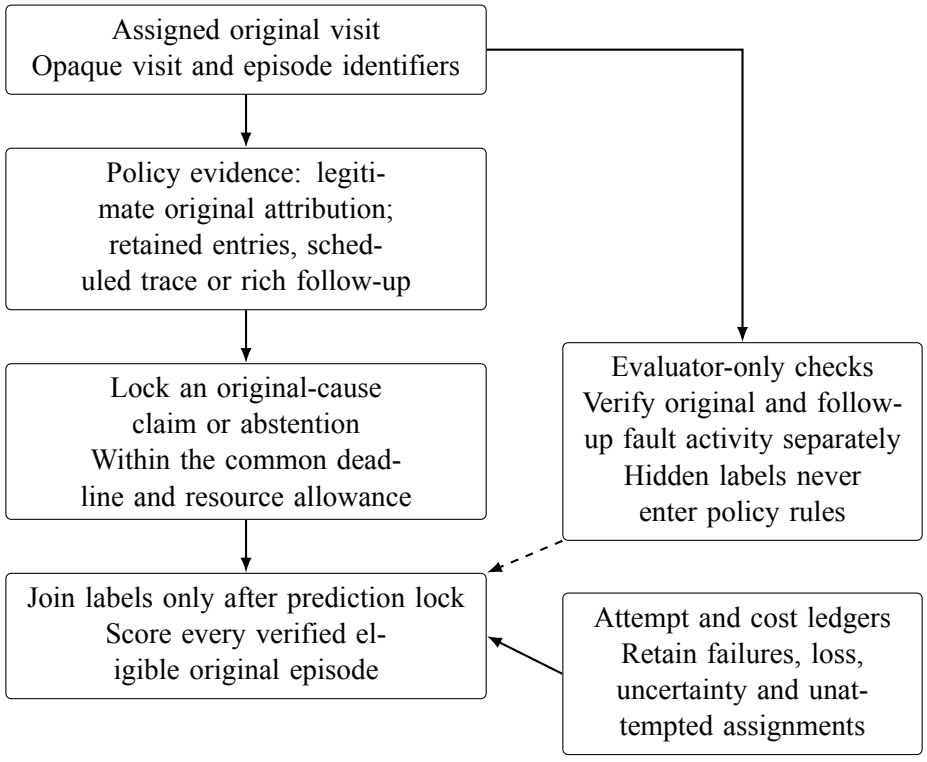


Figure 3.1: Proposed evidence and hidden-label flow for scoring the cause of the original episode.

C will retain at most 1,024 entries or 4 MiB of serialized data, whichever limit is reached first. Evictions and drops will be recorded with affected time ranges. At 15 seconds, its record will be stopped and frozen before alert finalization, then exported

Table 3.4: Proposed capture strategies. Adaptive tracing and retained original evidence are established mechanisms; their effectiveness and costs in this comparison remain to be measured (Darvishi et al. 2026; Zhang et al. 2023; Miller et al. 1995; Arnold and Ryder 2001; GoogleChrome/web-vitals contributors n.d.).

Arm	Evidence contract	Role
A	Metrics and all supported original-visit attribution, serialized before fields disappear.	Practical baseline.
B	A on the original visit; one rich follow-up after an alert, with A evidence retained.	Delayed diagnosis.
C	A plus an early record of supported Resource Timing, long-animation-frame and long-task entries; diagnostic access after alert.	Bounded original evidence.
D	A on every original; rich capture on every fourth assigned original per fixture, with randomized starting phase.	Periodic alternative.
E	A and rich capture on every original, under a separate reference allowance.	Coverage/cost reference.

within the visit deadline. Every C original, including an unalerted one, will be exported for conformance auditing to a separate, access-logged store inaccessible to the diagnosis worker. Every export will be charged. A diagnostic copy will be released only after the common alert; the classifier will have no early access through the audit store. Missing required earlier evidence will make the record incomplete. This limited entry record is not a full browser trace.

D's phase will be selected independently of the hidden fault schedule, and no additional visit will be introduced. E will neither provide labels nor share evidence with other arms. Its richer observation can itself perturb the workload or miss evidence, so it is not an oracle or guaranteed upper bound. Hindsight's bounded retention lifecycle motivates explicit horizon and loss checks, but its distributed-system capacities do not prescribe browser buffer sizes (Zhang et al. 2023; World Wide Web Consortium 2025).

3.4.2 Finite Allocation and Schedule

FP-0.1 specifies 64 matched blocks: two fixtures, eight scenarios and four repetitions. Each block will contain five separate originals, one per arm, giving 320 comparison originals. Each arm will receive 48 assigned fault opportunities, eight stable controls and eight nuisance controls. Transient and replaced-cause cases will each contribute 16 originals per arm. These are assigned counts, not guaranteed completed or verified observations.

Warm-up records will be retained but excluded from evaluation. Preparation and cal-

Table 3.5: Maximum planned browser starts in FP-0.1. All counts are feasibility allocations, not a power calculation.

Stage	Allocation	Starts
Warm-up	Two visits per fixture in each of three sessions	12
Stable history	Two fixtures, 40 A visits each	80
Qualification	Two fixtures, four conditions, M/C/rich contracts, five visits; four extra C checks per fixture	128
Perturbation calibration	Two fixtures, six blocks, four contracts	48
Comparison originals	Two fixtures, eight scenarios, four repetitions, five arms	320
B follow-ups	At most one per B original, including control alerts	64
Planned maximum	Preparation and comparison combined	652
Repair reserve	One pre-comparison repair cycle only	48
Absolute cap	No additional or replacement visits	700

ibration will occupy one session of at most three active hours. Two comparison sessions on different days will each have a 2.5-hour limit. Repetitions 1–2 will be allocated to the first comparison session and 3–4 to the second. Fixture/scenario blocks and arm order within each block will be shuffled. Seeds and schedule hashes will be saved before comparison, and the hidden fault schedule will be withheld from policies.

B will be assigned a target launch delay of 1 or 15 seconds after the original 15-second observation window, with each delay used twice across the four repetitions in a fixture/scenario cell. Actual delay to follow-up navigation, including startup and queueing, will be recorded. The common diagnosis deadline will be 90 seconds after the original observation window ends. Scheduling feasibility, especially the 1-second launch target alongside export and cleanup, will be checked before collection. Missed targets and late diagnoses will remain outcomes; further diagnostic visits will not be used to repair them.

Four repetitions allow both delays to be represented across two sessions; they do not establish population precision. The two fixtures remain two templates despite the larger number of visits. Published five-run Lighthouse recommendations and native-benchmark repetition tables will not be transferred as sample-size rules (Heričko, Šumak, and Brdnik 2021; Kalibera and Jones 2013).

3.4.3 Resource Accounting and Stops

The campaign will stop at the first of 700 starts, eight active hours, seven consecutive calendar days from the first warm-up or 8 GiB of artifacts. Active time includes startup, waits, resets, verification, capture and export. Gaps between sessions count towards the calendar limit. MiB and GiB denote binary storage units. Resident set size (RSS) will refer to memory resident in the browser and collector process tree.

Table 3.6: Proposed resource limits and responses. A–D have equal allowance vectors; E has a separate reference allowance.

Scope	Limit and response
Each visit	30 seconds elapsed, 30 CPU-seconds and 2 GiB peak process-tree RSS; a breach will terminate the visit and retain its reason.
Rich export	32 MiB per visit; truncation and loss will be marked.
C record	1,024 entries or 4 MiB serialized data; runtime memory will be measured separately. Serialized size is not heap usage.
Each A–D policy	Three charged hours, 120 CPU-minutes and 2 GiB cumulative exported evidence, with the common visit limits.
E reference	Three charged hours, 120 CPU-minutes and 3 GiB exported evidence, with the common visit limits.
Diagnosis and retries	Common 90-second deadline; at most one B follow-up; no comparison retries or visits until recurrence.

All original scans, retained or dropped records, serialization, exports, waits, confirmations, failed attempts and follow-ups will be charged. CPU time, elapsed time, memory and bytes will remain separate quantities. The full shared stable-history cost will be charged to each arm's standalone policy ledger, but its physical visits will be counted only once in the campaign ledger. Qualification and calibration will be reported as development costs outside policy allowances and inside campaign caps. Evaluator-only verification will remain hidden and will be costed separately.

After a policy exhausts its allowance, it will stop obtaining evidence. Remaining scheduled originals will still be executed with the minimally observed evaluator harness while campaign limits permit. Verified eligible episodes will remain in that policy's denominator as no correct in-time diagnosis. In contrast, a global cap or shared integrity stop will prevent remaining launches. Those cases will be marked unattempted without invented labels, and the campaign will be reported as incomplete. Evidence will not be deleted during collection to make a breached storage cap appear satisfied.

3.5 O4: Outcomes, Denominators and Analysis

3.5.1 Primary Comparison and Missing Labels

A correct attribution will name the independently verified fault class of the original episode within the common deadline and policy allowance. A follow-up-only explanation will be scored against the original label, even if the follow-up contains a different fault.

For arm s, let E_s contain its independently verified eligible original regression episodes and let $N_s = |E_s|$. Let c_s count episodes in E_s with a timely, within-budget, locked, non-abstaining prediction of the correct original class. The attribution rate and C-minus-B contrast will be

$$\widehat{p}_s = \frac{c_s}{N_s}, \qquad \widehat{\Delta}_{C-B} = \widehat{p}_C - \widehat{p}_B. \tag{3.1}$$

The principal contrast will use a prospectively defined intermittent subset. Pilot transient and replacement cases will be shown separately. A rate will be undefined when $N_s = 0$. Missed alerts, failed or late follow-ups, lost evidence, policy exhaustion and abstentions will remain in N_s as no correct diagnosis. Repeated alerts for one original episode will be deduplicated.

The five executions within a matched block can have different verified eligibility or unresolved-label patterns. Matching therefore does not imply equal observed denominators. The main-study analysis freeze will define the target population and site/scenario weighting, the treatment of unequal verified arm sets, and the sensitivity analysis for unresolved labels. No pairing or weighting method is assumed solely from the matched allocation. Pilot counts will be shown by arm and cell so that differential verification cannot disappear in a pooled rate.

For the pilot, both the 48 assigned fault opportunities per arm and the verified eligible denominator will be reported. An assignment-based sensitivity range will use known correct cases divided by 48 as its lower bound. Only attempted cases with unresolved eligibility or cause will be added to the numerator for the upper bound. Corresponding worst/best combinations will bound the descriptive C–B difference; transient and replacement subsets will use their own assigned denominators. These missing-label bounds are not confidence intervals and cannot turn a globally incomplete campaign into a completed comparison. Globally unattempted cases will remain separately identified.

3.5.2 Secondary Outcomes and Interpretation

Detection coverage will be the fraction of verified eligible episodes alerted. Diagnostic coverage will be the fraction receiving a timely, within-budget, non-abstaining cause claim. Conditional diagnostic precision will be correct original-cause claims divided by all such claims on verified eligible episodes; it will be undefined if no claims are made. Wrong original-cause claims will be reported over all verified eligi-

ble episodes with a fault-class confusion matrix. Coverage will accompany precision so that selective abstention cannot conceal missed episodes.

Stable and nuisance controls will be reported separately, with false-alert and unsupported cause-claim rates per observation and per unit of monitoring time. Abstention, timeout, failed follow-up, unsupported output, buffer loss, unavailable evidence and exhausted allowance will be recorded with overlapping reason codes where necessary. Assigned and actual follow-up delay, independently checked recurrence and diagnosis time will be reported. Unresolved episodes will remain visible alongside the timing summaries of successful diagnoses.

Costs will be reported in total, per scheduled original and per correctly attributed episode. Cost per correct attribution will be undefined when no episode is correctly attributed. Counts and rates will be broken down by fixture, scenario, arm and session. Pilot summaries will include medians, ranges, variation across blocks and sessions, and ordering or drift plots. No efficacy significance test or two-fixture cluster bootstrap will be presented as population uncertainty.

The main study will require a practical effect or precision target, justified independent-unit coverage, a sample-size rationale, an appraised interval or test, multiplicity handling, missing-label analysis and a stopping rule. Pilot variation and costs will inform a range of sample-size scenarios rather than a calculation based on one favorable effect. Nested benchmarking and performance-fault studies motivate this care, but do not supply the proposed study's sample size or causal oracle (Kalibera and Jones 2013; Kalibera and Jones 2020; Delgado-Pérez et al. 2021). An inconclusive comparison will not establish equivalence.

3.6 O5: Reliability Boundaries and Validity

Fault persistence will be the principal factor. Attribution and cost will be compared across persistent, transient and replaced-cause conditions. Follow-up delay and controlled browser-state changes will be secondary factors where the budget and later protocol support them. A condition-by-strategy summary will distinguish adequate attribution, evidence loss and inconclusive results using a practical criterion fixed before main-study outcomes are inspected.

The pilot will use a fresh browser process and empty profile, cleared fixture/controller state and disabled service workers. Host load and previous-run state will be recorded; complete host-cache clearing will not be assumed. A later browser-state experiment would require one-factor contrasts where feasible. Uncontrolled changes in several conditions would not establish an isolated cache effect (Netravali et al. 2018; Barrett et al. 2017; Microsoft / Playwright contributors n.d.; Zeber et al. 2020).

Internal-validity risks include label leakage, incorrect visit linkage, missing entries, measurement perturbation, state carryover and tuning on evaluation data. Separate evaluator access, opaque identifiers, independent activity checks, randomized order, versioned manifests and complete attempt records will address these risks. Repeated

visits and related fault occurrences will retain their dependence identifiers. The small number of fixture templates limits external validity; localhost conditions cannot establish production fault prevalence or real-user accuracy (Grassi et al. 2021; Demir et al. 2022; Jueckstock, Sarker, et al. 2021). Any later authorized live-site sample will assess operational robustness, not supply verified fault truth from unexplained slowdowns.

3.6.1 Data Ethics, Retention and Reproducibility

The pilot will use owned localhost fixtures, fixed safe inputs and no human participants or real-user traces. Credentials and unnecessary page or user content will not be stored. Policy evidence and evaluator labels will use separate access paths, and an append-only manifest will preserve attempts, hashes, loss codes and deviations. Access logging will support the audit of C's retained evidence and hidden labels.

Within the 8-GiB cap, raw pilot artifacts will be retained for 90 days after pilot-report sign-off. The de-identified ledger, configurations, derived tables and hashes will be retained through project assessment. Actual storage and deletion actions will be recorded when due. Any later live-site work will require its own authorization, access conditions and data-minimization rules.

3.7 Pilot Progression and Main-Study Separation

Before collection, the execution manifest will be completed with the host, OS, CPU, RAM, browser and automation builds, package checksums, fixture commits, capture categories, clock alignment, diagnosis mapping, seeds, storage location, start date and accountable researcher. Available host capacity and enforcement of the caps will be checked. Qualification will include four additional C checks per fixture: buffer overflow, export failure, a trigger delayed to 20 seconds and absence of the qualifying input. Expected failures in these checks will be separated from unexpected failures in ordinary visits.

The proposed progression rules are engineering decisions, not statistical acceptance tests. For each fixture and class under M, all five qualification injections must execute, at least four must exceed the alert threshold, and median LCP must increase by at least 200 ms relative to stable M visits. Fault activity under C and rich capture will also be checked. C must supply correctly scoped, exportable resource and main-thread discriminators in every ordinary qualification case; its frozen rules must distinguish each class in at least four of five cases per fixture. Rich-capture discriminator mapping will be checked separately.

No required-record loss is permitted for ordinary C qualification visits, and deliberately induced loss must be detected. Stable median LCP increases for both A and C relative to M must be no greater than the larger of 25 ms and 10% of the M median for each fixture, without altered fault activity. Increases above this level but no greater

than the larger of 60 ms and 25%, or unstable estimates, will require amendment; larger changes or altered mechanisms will stop this protocol version. Rich-reference perturbation will be reported separately.

Operational progression after comparison requires every design cell to have been attempted, at least 58 of 64 originals per arm to have completed with valid exported evidence, and at least 46 of 48 assigned fault labels to have been verified. Counts of at least 48 of 64 and 39 of 48 respectively may support amendment if a specific feasible correction exists. Lower counts, an unattempted cell or an unworkable resource requirement will stop this version. Every known-cause score must have valid independent verification and linkage. Label leakage, wrong visit linkage or unreliable shared records will cause an immediate integrity stop.

Green progression means that a main-study protocol may be prepared; it does not establish efficacy or authorize automatic main collection. Overall amber requires a specific prospective amendment. This differs from the single repair cycle allowed *before* comparison: at most 48 extra starts within all campaign caps, with a documented failure and correction. The only permitted severity increase is from 500 to 1,000 ms if qualification fails to produce the intended symptom. A new fault class, fixture, detector, capture mechanism or larger budget requires a new protocol version. Comparison outcomes will not be used to tune that same comparison.

A signalled arm-specific timeout, overflow or missing record will remain an outcome; other scheduled work may continue while shared integrity and caps permit. If A already diagnoses the selected causes, that result will be retained. If C cannot provide useful original evidence, its scope or the research comparison will be revised prospectively. C outperforming B is not a progression requirement.

The pilot readout is limited to six analyst-hours within two calendar days of the collection stop. It will contain run and cost ledgers, fixture and capture manifests, conformance and loss checks, per-cell outcome tables, perturbation summaries, deviations and a reasoned stop, amend or prepare-main-study decision. An unfinished readout will be recorded as incomplete without extending collection. Main-study collection will depend on a separate protocol and analysis freeze, and all pilot and repair data will remain outside the confirmatory dataset.

04. RESEARCH WORK PLAN

4.1 Sequence and Dependencies

The research will proceed from measurement and fault verification to a bounded pilot, followed by a separate main-study decision. Collection will depend on verified host capacity, fixed software and fixture versions, isolated evaluator labels, qualified capture contracts and enforceable resource limits. Main-study collection will additionally depend on a reviewed protocol that defines the target population, allocation, outcomes, analysis and stopping rule.

Table 4.1 gives relative stages rather than unconfirmed calendar dates. The person-day ranges are provisional planning estimates. FP-0.1's seven-calendar-day and eight-active-hour limits are protocol caps, distinct from those effort estimates. The researcher will be responsible for implementation, collection, records and analysis. The main protocol and progression decision will require review; the reviewer has not yet been assigned.

Table 4.1: Research milestones, provisional effort and completion evidence.

Stage	Research activity	Effort or limit	Dependency and deliverable
R1	Complete the focused related-work comparison	3–5 person-days	Claims will be tied to inspected source editions. Any additional method adopted will require an appraisal of its assumptions and relevance. The output will be a bounded contribution statement.
R2	Implement measurement contracts, fixtures and oracle	5–8 person-days	Fixed builds, exact capture fields, state-reset procedures, hidden activity checks and linked cost records will be produced. Both faults must be supported by both fixtures.
R3	Establish execution readiness	2–4 person-days	Host capacity, watchdogs, clock alignment, access separation, classifier mapping and schedule will be checked. The completed execution manifest will be required before pilot collection.

Continued on next page

Stage	Research activity	Effort or limit	Dependency and deliverable
R4	Conduct conditional FP-0.1	At most 7 calendar days and 8 active hours	The frozen allocation will be run within 652 planned starts, a 700-start absolute cap and 8 GiB. Every attempt, failure and stop will be recorded.
R5	Assess feasibility and define the main protocol	3–6 person-days; pilot readout within its separate cap	The initial readout is limited to 6 analyst-hours within 2 calendar days of collection stop. A reasoned stop, amend or prepare-main decision will precede a reviewed main-study protocol.
R6	Conduct conditional main collection	Set after R5	A justified sample, independent-unit coverage, resource allocation and frozen analysis will be required. Locked predictions, all-attempt records and deviations will form the evaluation dataset.
R7	Analyse outcomes and reliability conditions	Set after R5	Original-cause attribution, uncertainty, failures, complete costs and condition-specific limits will be reported under the frozen analysis. Confirmatory and exploratory findings will be distinguished.

The stages have scientific dependencies. Capture contracts cannot be qualified without a reproducible fixture and independent execution check. Strategy costs cannot be compared without both the physical campaign ledger and the standalone policy ledgers. A main-study sample cannot be justified by the pilot visit count alone; variation, dependence, independent fixture coverage and a practical effect or precision target must first be considered (Kalibera and Jones 2013; Kalibera and Jones 2020).

4.2 Resources and Deliverables

The initial resources will be a localhost test environment, a host able to enforce the proposed CPU and memory limits, one fixed Chromium and automation stack, isolated evaluator storage, access logs and a watchdog for time, memory and storage limits. Resource availability will be verified before the schedule begins. The pilot's caps will limit feasibility work; any later main-study budget will be set from measured costs and the chosen design.

The research deliverables will be a versioned measurement protocol and data dictionary; reproducible fixtures and independent fault checks; a longitudinal episode

dataset and complete attempt ledger; capture configurations and cost records for A–E; conformance and perturbation evidence; and a comparison of diagnostic outcomes and reliability conditions. A stopped or amended pilot will still produce an auditable record of what was attempted and why the protocol could not proceed.

The related engineering work will provide the browser profiler service/API and seven performance views described in Chapter 1. It will include metric, network, main-thread, memory and timeline outputs, reference-tool comparisons and the specified unit tests. This delivery can use the research contracts and provenance structure, but its broader feature coverage will remain separate from the pilot's LCP-based causal comparison. Reference-tool agreement will be reported as measurement consistency, not independent proof of the original fault.

4.3 Progression, Change Control and Risks

The progression rules in Section 3.7 will determine whether the pilot version is stopped, amended or used to prepare the main protocol. A green outcome will indicate only that further study design is justified. An amber outcome will require a specific prospective amendment. The one permitted pre-comparison repair cycle will remain within its 48-start reserve and all campaign caps. A change to the fault class, fixture, detector, capture mechanism or resource allocation will require a new protocol version.

After comparison begins, outcomes will not be used to adjust that same comparison. Detected arm-specific loss or timeout will remain an outcome, while shared record-integrity failures will end the version. Any later change to the main protocol will be documented and reviewed before the affected collection. This sequence preserves the distinction between feasibility learning and confirmatory evaluation.

Table 4.2: Main research risks and planned responses.

Risk	Planned control	Response
Label leakage or wrong linkage	Separate evaluator access, opaque identifiers, per-visit checks and hashes.	Shared integrity failure will stop the campaign; partial records will be retained.
Missing original evidence	Early collection, loss checks, access audit and export-cost accounting.	Affected opportunities will remain failures; an infeasible comparator will require prospective revision.
Instrumentation alters the fault	Randomized calibration against minimal observation.	Bounded repair before comparison, or termination of this protocol version.
Follow-up sees another cause	Separate visit identities and independent recurrence checks.	The diagnosis will be scored against the original fault and transfer errors reported.
Dependence or narrow coverage	Preserve template, occurrence, session, block and visit identifiers.	Inference will be limited; main-study coverage and analysis will be justified separately.
Campaign cap reached	Watchdog and complete time, start-count and storage records.	Remaining cases will be unattempted, and the campaign will be reported as incomplete.

05. CONCLUSION

A detailed profile from a later browser visit may explain that execution without identifying the cause of an earlier regression. The proposed research addresses this distinction by comparing follow-up profiling with diagnostic evidence retained during the original visit. Its central question concerns correct attribution of the original cause under stated resource allowances and a common deadline, rather than whether a later visit is also slow.

The five objectives connect repeatable measurement, a longitudinal dataset, independent fault labels, capture-strategy comparison and an assessment of reliability conditions. Persistent, transient and replaced-cause scenarios will distinguish recurrence from symptom similarity. Lightweight attribution will remain a strong baseline, and follow-up profiling will retain that original evidence. Bounded original capture and periodic capture will be evaluated alongside it, while always-rich capture will provide a separately costed reference. Missed alerts, failed captures, late answers, abstentions and policy exhaustion will remain in the verified eligible episode denominator.

The finite pilot will first establish whether fault checks, capture contracts, scheduling and complete resource accounting support the comparison. Its two fixture templates and repeated visits are a feasibility allocation, not a powered efficacy study or a representative sample of production websites. Any main experiment will require a separate protocol and analysis freeze, with pilot and repair data excluded from confirmatory evaluation. Existing work already establishes adaptive instrumentation, selective tracing and retention of original execution evidence, so the intended contribution is a bounded reliability and cost assessment rather than a claim that these mechanisms are new.

Neither the pilot nor the main study has been completed. No diagnostic advantage, hypothesis result or final novelty claim can therefore be established from this report. The next research milestone is a verified execution manifest and qualified measurement, fault and evidence contracts. If feasibility is established, the study can provide an evidence-based account of when delayed profiling is useful, when retained evidence adds value and when attribution should remain uncertain. Its conclusions will be limited to the fixtures, faults, browser versions, states and resource conditions actually tested.

REFERENCES

[1] *[VIM3] 2.13 measurement accuracy* [Online] (n.d.). Available at: https://jcgm.bipm.org/vim/en/2.13.html [Accessed: 10/04/2026].

[2] *[VIM3] 2.18 measurement bias* [Online] (n.d.). Available at: https://jcgm.bipm.org/vim/en/2.18.html [Accessed: 10/04/2026].

[3] *5.3.3.2. Randomized block designs* [Online] (n.d.). Available at: https://www.itl.nist.gov/div898/handbook/pri/section3/pri332.htm [Accessed: 10/04/2026].

[4] Arnold, M. and B. G. Ryder (May 1, 2001). "A framework for reducing the cost of instrumented code". In: *Proceedings of the ACM SIGPLAN 2001 conference on Programming language design and implementation*. PLDI '01. New York, NY, USA: Association for Computing Machinery, pp. 168–179. DOI: 10.1145/378795.378832. URL: https://dl.acm.org/doi/10.1145/378795.378832 [Accessed: 08/29/2026].

[5] Barrett, E., C. F. Bolz-Tereick, R. Killick, S. Mount, and L. Tratt (October 12, 2017). "Virtual Machine Warmup Blows Hot and Cold". In: *Proceedings of the ACM on Programming Languages* 1 (OOPSLA), pp. 1–27. DOI: 10.1145/3133876. arXiv: 1602.00602[cs.PL]. URL: http://arxiv.org/abs/1602.00602 [Accessed: 10/04/2026].

[6] *Chrome DevTools Protocol* [Online] (n.d.). Available at: https://chromedevtools.github.io/devtools-protocol/tot/Network/ [Accessed: 08/29/2026].

[7] Chromium authors (n.d.). *Chrome DevTools Protocol: Tracing domain* [Online]. Tip-of-tree protocol documentation. Available at: https://chromedevtools.github.io/devtools-protocol/tot/Tracing/ [Accessed: 10/31/2026].

[8] *Cumulative Layout Shift (CLS) | Articles* [Online] (n.d.). web.dev. Available at: https://web.dev/articles/cls [Accessed: 08/29/2026].

[9] Darvishi, K., M. Noferesti, Y. Sehgal, and N. Ezzati-Jivan (2026). "LMAT: An adaptive tracing approach based on efficient system behavior analysis using language models". In: *Journal of Systems and Software* 238, p. 112890. DOI: 10.1016/j.jss.2026.112890. URL: https://doi.org/10.1016/j.jss.2026.112890 [Accessed: 10/31/2026].

[10] Delgado-Pérez, P., A. B. Sánchez, S. Segura, and I. Medina-Bulo (2021). “Performance mutation testing”. In: *Software Testing, Verification and Reliability* 31.5, e1728. DOI: 10.1002/stvr.1728. URL: https://onlinelibrary.wiley.com/doi/abs/10.1002/stvr.1728 [Accessed: 10/04/2026].

[11] Demir, N., M. Große-Kampmann, T. Urban, C. Wressnegger, T. Holz, and N. Pohlmann (April 25, 2022). “Reproducibility and Replicability of Web Measurement Studies”. In: *Proceedings of the ACM Web Conference 2022*. WWW ’22: The ACM Web Conference 2022. Virtual Event, Lyon France: ACM, pp. 533–544. DOI: 10.1145/3485447.3512214. URL: https://dl.acm.org/doi/10.1145/3485447.3512214 [Accessed: 10/04/2026].

[12] *Emulation - DevTools Protocol* [Online] (n.d.). Available at: https://chromedevtools.github.io/devtools-protocol/#/Emulation [Accessed: 10/04/2026].

[13] *Event Timing API* [Online] (n.d.). Available at: https://www.w3.org/TR/event-timing/ [Accessed: 08/29/2026].

[14] GoogleChrome/web-vitals contributors (n.d.). *web-vitals library: usage and attribution build* [Online]. Version 5.1.0. Available at: https://github.com/GoogleChrome/web-vitals/blob/v5.1.0/README.md [Accessed: 10/31/2026].

[15] Grassi, G., R. Teixeira, C. Barakat, and M. Crovella (2021). “Leveraging Website Popularity Differences to Identify Performance Anomalies”. In: *IEEE INFOCOM 2021 - IEEE Conference on Computer Communications*. IEEE. DOI: 10.1109/INFOCOM42981.2021.9488832. URL: https://www.cs.bu.edu/fac/crovella/paper-archive/infocom21-website-anomalies.pdf [Accessed: 10/31/2026].

[16] Hantke, F., P. Snyder, H. Haddadi, and B. Stock (June 4, 2025). *Web Execution Bundles: Reproducible, Accurate, and Archivable Web Measurements*. DOI: 10.48550/arXiv.2501.15911. arXiv: 2501.15911v4. URL: https://arxiv.org/abs/2501.15911v4 [Accessed: 10/04/2026].

[17] Heričko, T., B. Šumak, and S. Brdnik (2021). “Towards Representative Web Performance Measurements with Google Lighthouse”. In: *Proceedings of the 2021 7th Student Computer Science Research Conference (StuCoSReC)*. University of Maribor Press, pp. 39–42. DOI: 10.18690/978-961-286-516-0.9. URL: https://doi.org/10.18690/978-961-286-516-0.9.

[18] *HTTP* [Online] (n.d.). *HTTP: Primer on Web Performance - High Performance Browser Networking (O’Reilly)* [Online]. High Performance Browser Networking.

Available at: https://hpbn.co/primer-on-web-performance/ [Accessed: 10/04/2026].

[19] *Interaction to Next Paint (INP) | web.dev* [Online] (n.d.). Available at: https://web.dev/articles/inp [Accessed: 08/29/2026].

[20] Jueckstock, J. and A. Kapravelos (October 21, 2019). "VisibleV8: In-browser Monitoring of JavaScript in the Wild". In: *Proceedings of the Internet Measurement Conference*. IMC '19. New York, NY, USA: Association for Computing Machinery, pp. 393–405. DOI: 10.1145/3355369.3355599. URL: https://dl.acm.org/doi/10.1145/3355369.3355599 [Accessed: 08/29/2026].

[21] Jueckstock, J., S. Sarker, P. Snyder, A. Beggs, P. Papadopoulos, M. Varvello, B. Livshits, and A. Kapravelos (April 19, 2021). "Towards Realistic and ReproducibleWeb Crawl Measurements". In: *Proceedings of the Web Conference 2021*. WWW '21: The Web Conference 2021. Ljubljana Slovenia: ACM, pp. 80–91. DOI: 10.1145/3442381.3450050. URL: https://dl.acm.org/doi/10.1145/3442381.3450050 [Accessed: 10/04/2026].

[22] Kalibera, T. and R. Jones (June 20, 2013). "Rigorous benchmarking in reasonable time". In: *ACM SIGPLAN Notices* 48.11, pp. 63–74. DOI: 10.1145/2555670.2464160. URL: https://dl.acm.org/doi/10.1145/2555670.2464160 [Accessed: 10/04/2026].

[23] Kalibera, T. and R. Jones (July 21, 2020). *Quantifying Performance Changes with Effect Size Confidence Intervals*. DOI: 10.48550/arXiv.2007.10899. arXiv: 2007.10899[stat.ME]. URL: http://arxiv.org/abs/2007.10899 [Accessed: 10/04/2026].

[24] *Largest Contentful Paint | Lighthouse* [Online] (n.d.). Chrome for Developers. Available at: https://developer.chrome.com/docs/lighthouse/performance/lighthouse-largest-contentful-paint [Accessed: 10/04/2026].

[25] *Layout Instability API* [Online] (n.d.). Available at: https://wicg.github.io/layout-instability/ [Accessed: 08/29/2026].

[26] *Long Animation Frames API* [Online] (n.d.). Available at: https://www.w3.org/TR/long-animation-frames/ [Accessed: 08/29/2026].

[27] *Long Tasks API* [Online] (n.d.). Available at: https://www.w3.org/TR/longtasks-1/ [Accessed: 08/29/2026].

[28] Mertz, J. and I. Nunes (2023). "Software runtime monitoring with adaptive sampling rate to collect representative samples of execution traces". In: *Journal of Systems and Software* 202. Author preprint: https://arxiv.org/abs/2305.

01039v1 (1 May 2023), p. 111708. DOI: 10.1016/j.jss.2023.111708. URL: https://doi.org/10.1016/j.jss.2023.111708 [Accessed: 10/31/2026].

[29] Microsoft / Playwright contributors (n.d.). *Playwright: Isolation and browser contexts* [Online]. Available at: https://playwright.dev/docs/browser-contexts [Accessed: 10/31/2026].

[30] Miller, B. P., M. D. Callaghan, J. M. Cargille, J. K. Hollingsworth, R. B. Irvin, K. L. Karavanic, K. Kunchithapadam, and T. Newhall (1995). *The Paradyn Parallel Performance Measurement Tools*. Project manuscript. URL: https://ftp.cs.wisc.edu/paradyn/papers/Miller95Paradyn.pdf [Accessed: 08/29/2026].

[31] Mytkowicz, T., A. Diwan, M. Hauswirth, and P. F. Sweeney (March 7, 2009). "Producing wrong data without doing anything obviously wrong!" In: *Proceedings of the 14th international conference on Architectural support for programming languages and operating systems*. ASPLOS XIV. New York, NY, USA: Association for Computing Machinery, pp. 265–276. DOI: 10.1145/1508244.1508275. URL: https://dl.acm.org/doi/10.1145/1508244.1508275 [Accessed: 08/29/2026].

[32] Netravali, R., V. Nathan, J. Mickens, and H. Balakrishnan (2018). "Vesper: Measuring Time-to-Interactivity for Web Pages". In: *15th USENIX Symposium on Networked Systems Design and Implementation (NSDI 18)*. USENIX Association, pp. 217–231. URL: https://www.usenix.org/conference/nsdi18/presentation/netravali-vesper [Accessed: 08/29/2026].

[33] *Performance panel* [Online] (n.d.). *Performance panel: Analyze your website's performance | Chrome DevTools* [Online]. Chrome for Developers. Available at: https://developer.chrome.com/docs/devtools/performance/overview [Accessed: 10/04/2026].

[34] *Resource Timing* [Online] (n.d.). Available at: https://www.w3.org/TR/resource-timing/ [Accessed: 08/29/2026].

[35] Sambasivan, R. R., A. X. Zheng, M. De Rosa, E. Krevat, S. Whitman, M. Stroucken, W. Wang, L. Xu, and G. R. Ganger (2011). "Diagnosing Performance Changes by Comparing Request Flows". In: *8th USENIX Symposium on Networked Systems Design and Implementation (NSDI 11)*. USENIX Association. URL: https://www.usenix.org/conference/nsdi11/diagnosing-performance-changes-comparing-request-flows [Accessed: 10/31/2026].

[36] Shahedi, K., H. Li, M. Lamothe, and F. Khomh (2024). *Tracing Optimization for Performance Modeling and Regression Detection*. arXiv:2411.17548v1.

Preprint, version 1, 26 November 2024. arXiv. URL: https://arxiv.org/abs/2411.17548v1 [Accessed: 10/31/2026].

[37] *Speed Index | Lighthouse* [Online] (n.d.). Chrome for Developers. Available at: https://developer.chrome.com/docs/lighthouse/performance/speed-index [Accessed: 08/29/2026].

[38] *Time to Interactive | Lighthouse* [Online] (n.d.). Chrome for Developers. Available at: https://developer.chrome.com/docs/lighthouse/performance/interactive [Accessed: 10/04/2026].

[39] *Total Blocking Time | Lighthouse* [Online] (n.d.). Chrome for Developers. Available at: https://developer.chrome.com/docs/lighthouse/performance/lighthouse-total-blocking-time [Accessed: 10/04/2026].

[40] Wang, X. S., A. Balasubramanian, A. Krishnamurthy, and D. Wetherall (2013). "Demystifying Page Load Performance with WProf". In: *10th USENIX Symposium on Networked Systems Design and Implementation (NSDI 13)*. USENIX Association, pp. 473–485. URL: https://www.usenix.org/conference/nsdi13/technical-sessions/presentation/wang.

[41] Web Platform Incubator Community Group (n.d.). *JS Self-Profiling API draft* [Online]. Available at: https://wicg.github.io/js-self-profiling/ [Accessed: 10/31/2026].

[42] Weng, L., P. Huang, J. Nieh, and J. Yang (2021). "Argus: Debugging Performance Issues in Modern Desktop Applications with Annotated Causal Tracing". In: *2021 USENIX Annual Technical Conference (USENIX ATC 21)*. USENIX Association, pp. 193–207. URL: https://www.usenix.org/conference/atc21/presentation/weng [Accessed: 10/31/2026].

[43] *What's new in DevTools, Chrome 143 | Blog* [Online] (n.d.). Chrome for Developers. Available at: https://developer.chrome.com/blog/new-in-devtools-143 [Accessed: 10/04/2026].

[44] World Wide Web Consortium (May 21, 2025). *Performance Timeline* [Online]. Candidate Recommendation Draft, 21 May 2025. Available at: https://www.w3.org/TR/2025/CRD-performance-timeline-20250521/ [Accessed: 11/01/2026].

[45] World Wide Web Consortium (August 26, 2026a). *Largest Contentful Paint* [Online]. Working Draft, 26 August 2026. Available at: https://www.w3.org/TR/2026/WD-largest-contentful-paint-20260826/ [Accessed: 11/01/2026].

[46] World Wide Web Consortium (September 1, 2026b). *Navigation Timing Level 2* [Online]. Working Draft, 1 September 2026. Available at: https://www.w3.org/TR/2026/WD-navigation-timing-2-20260901/ [Accessed: 10/31/2026].

[47] World Wide Web Consortium (October 2, 2026c). *Paint Timing* [Online]. Working Draft, 2 October 2026. Available at: https://www.w3.org/TR/2026/WD-paint-timing-20261002/ [Accessed: 10/31/2026].

[48] Zeber, D., S. Bird, C. Oliveira, W. Rudametkin, I. Segall, F. Wollsén, and M. Lopatka (April 20, 2020). "The Representativeness of Automated Web Crawls as a Surrogate for Human Browsing". In: *Proceedings of The Web Conference 2020*. WWW '20. New York, NY, USA: Association for Computing Machinery, pp. 167–178. DOI: 10.1145/3366423.3380104. URL: https://dl.acm.org/doi/10.1145/3366423.3380104 [Accessed: 10/04/2026].

[49] Zhang, L., Z. Xie, V. Anand, Y. Vigfusson, and J. Mace (2023). "The Benefit of Hindsight: Tracing Edge-Cases in Distributed Systems". In: *20th USENIX Symposium on Networked Systems Design and Implementation (NSDI 23)*. USENIX Association, pp. 321–339. URL: https://www.usenix.org/conference/nsdi23/presentation/zhang-lei [Accessed: 10/31/2026].